When we run listing 3.6 on our trained model (trained for 5,000 epochs), we get about 90% accuracy. Your accuracy may be slightly better or worse. This certainly suggests it has learned *something* about how to play the game, but it's not exactly what we would expect if the algorithm really knew what it was doing (although you could probably improve the accuracy with a much longer training time). Once you actually know how to play, you should be able to win every single game.

There's a small caveat that some of the initialized games may actually be impossible to win, so the win percentage may never reach 100%; there is no logic preventing the goal from being in the corner, stuck behind a wall and pit, making the game unwinnable. The Gridworld game engine does prevent most of the impossible board configurations, but a small number can still get through. Not only does this mean we can't win every game, but it also means the learning will be mildly corrupted, since it will attempt to follow a strategy that normally would work but fails for an unwinnable game. We wanted to keep the game logic simple to focus on illustrating the concepts so we did not program in the sophisticated logic needed to ensure 100% winnable games.

There's also another reason we're being held back from getting into the 95% + accuracy territory. Let's look at our loss plot, shown in figure 3.14 showing our running average loss (yours may vary significantly).

Figure 3.14. The DQN loss plot after implementing experience replay, which shows a clearly down-trending loss, but it's still very noisy.